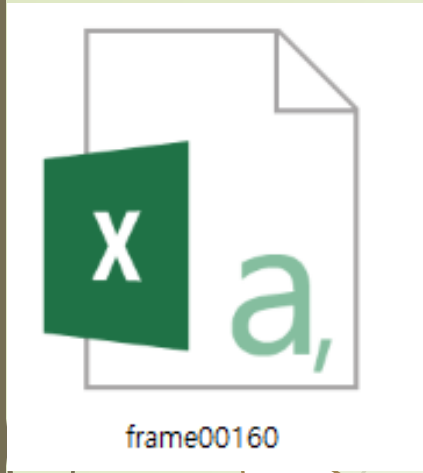
Source of Heat

Ho jun, Yang

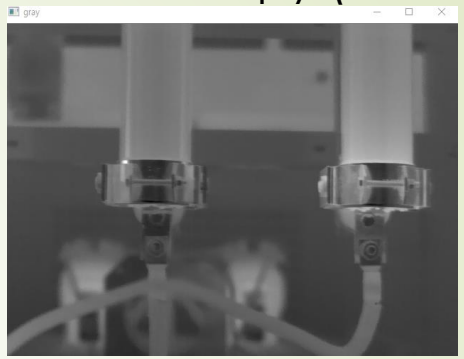


CSV file

.70552	36.70552	36.70552	36.7055
.71791	36.71791	36.71791	36.7179
.71172	36.71172	36.71172	36.7117
.71172	36.71172	36.71172	36.6559
.69312	36.69312	36.69312	36.6931
.71172	36.65593	36.68693	36.6869
.66833	36.66833	36.66833	36.6683

Normalize

Out of the Numpy (640 x 480)

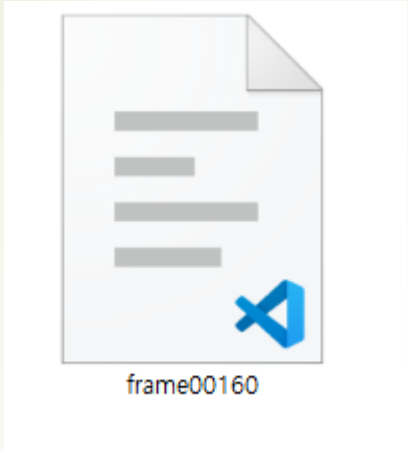


Numpy

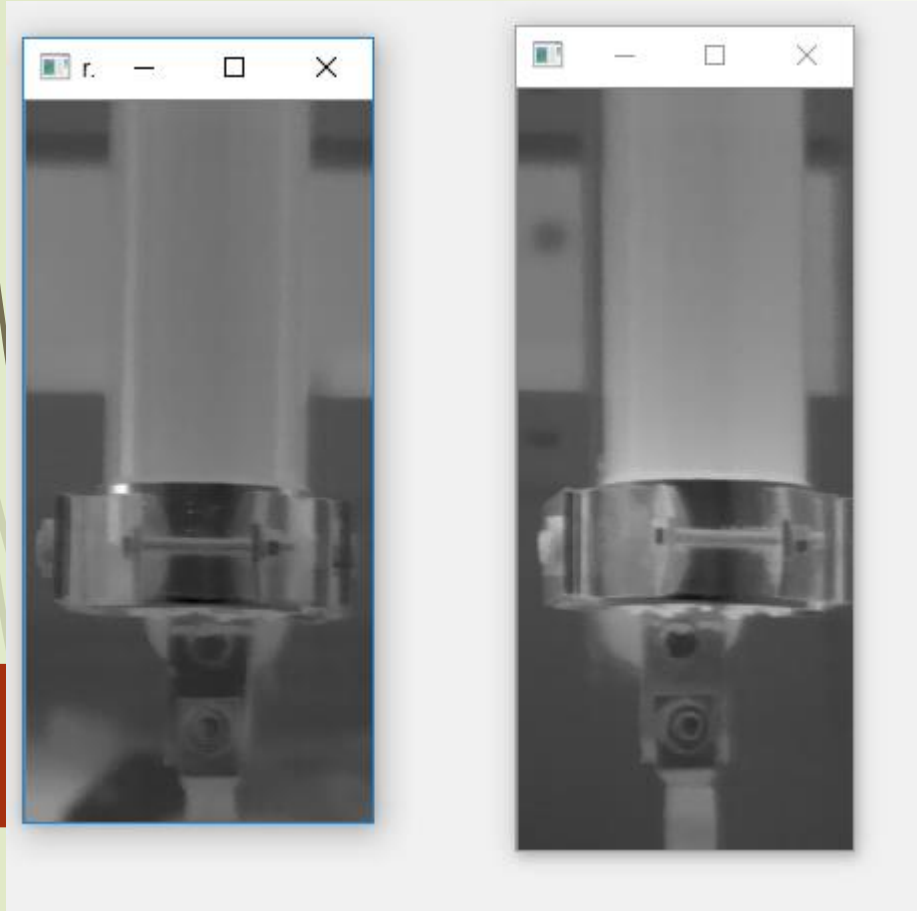
87	92	91	95	95	98
88	91	93	94	97	97
87	90	92	93	95	101
86	89	93	93	95	100
84	86	90	92	95	99
83	85	86	91	93	95
80	83	82	85	89	91
77	78	78	81	82	84
70	70	71	71	76	76
64	64	63	62	65	65
63	61	60	59	60	61
64	65	62	62	61	60
70	67	68	67	67	67
78	77	77	76	76	75
86	87	86	86	86	86

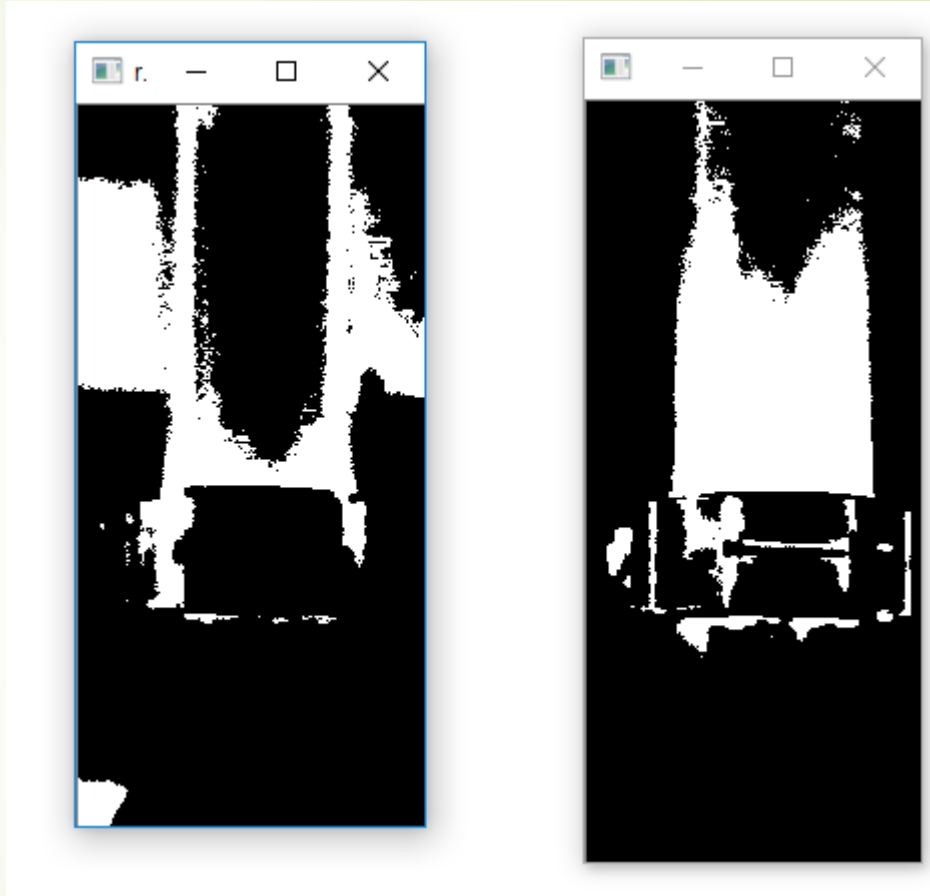
XML Data(Label Coordinate)

87	92	91	95	95	98
88	91	93	94	97	97
87	90	92	93	95	101
86	89	93	93	95	100
84	86	90	92	95	99
83	85	86	91	93	95
80	83	82	85	89	91
77	78	78	81	82	84
70	70	71	71	76	76
64	64	63	62	65	65
63	61	60	59	60	61
64	65	62	62	61	60
70	67	68	67	67	67
78	77	77	76	76	75
86	87	86	86	86	86



Apply Quartile Algorithm

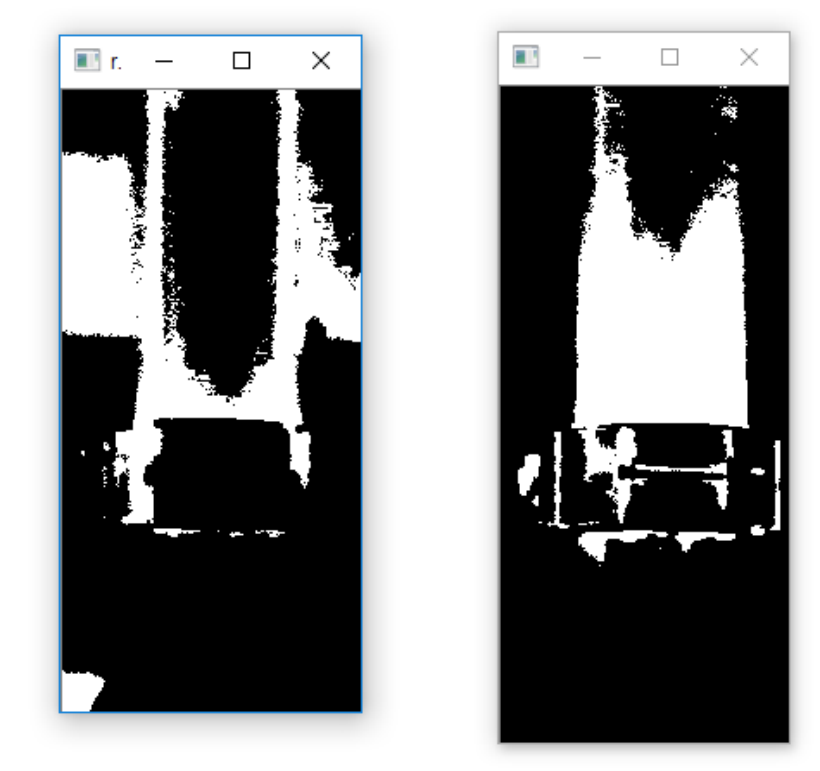




Data set : 7, 9, 16, 36, 39, 45, 45, 46, 48, 51

Q1 = 14.25 (25%)
Q2 = 42 (50%)
Q3 = 46.5 (75%)

Apply BFS(Breath-First Search) Algorithm



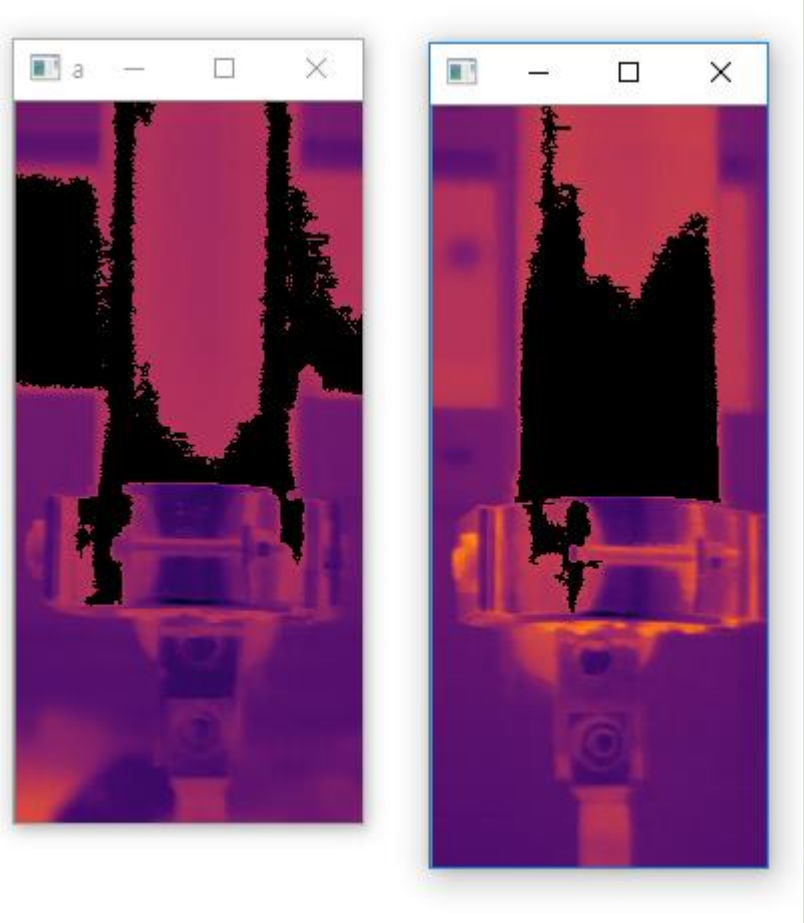
Reverse images and merge with png file

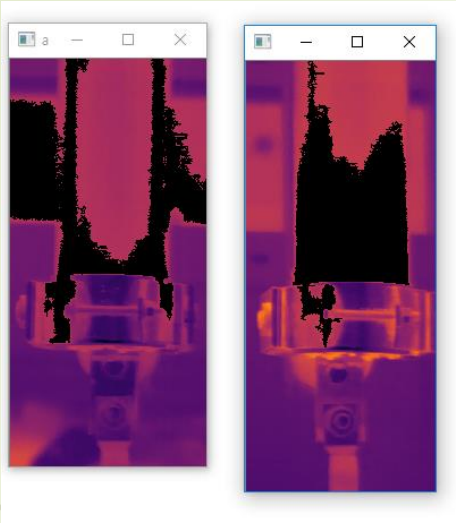
Pseudocode

Input: A graph *Graph* and a *starting vertex root* of *Graph*

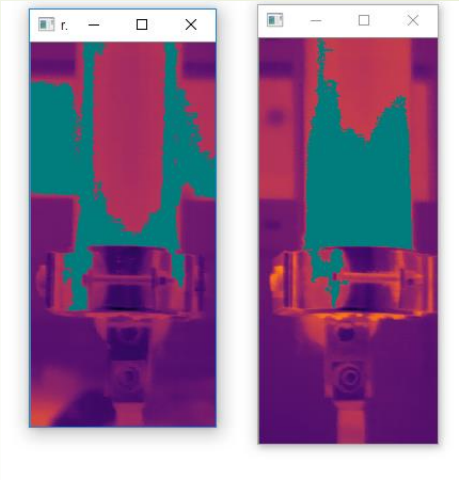
Output: Goal state. The *parent* links trace the shortest path back to *root*

```
procedure BFS(G, start_v):
    let Q be a queue
    label start_v as discovered
    Q.enqueue(start_v)
    while Q is not empty
        v = Q.dequeue()
        if v is the goal:
            return v
        for all edges from v to w in G.adjacentEdges(v) do
            if w is not labeled as discovered:
                label w as discovered
                w.parent = v
                Q.enqueue(w)
```





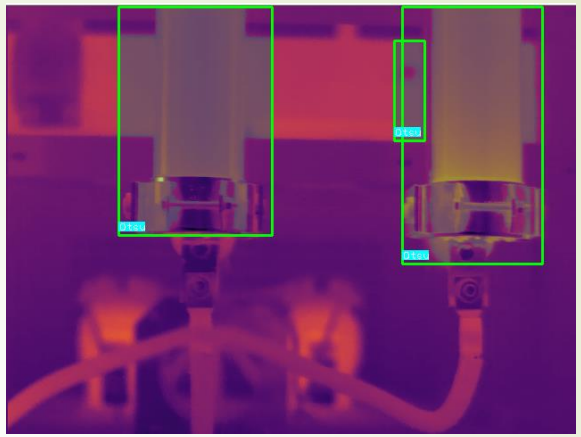
Change color of black pixels



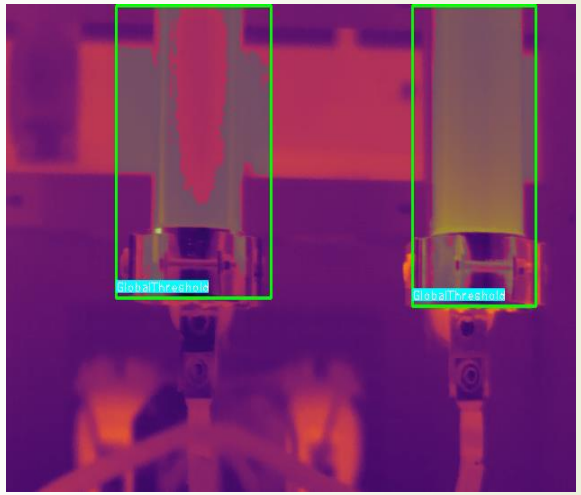
```
array([14665,   124,     1,   296,   252])
```

```
[array([13240,   474,     1,   617,   253])
```

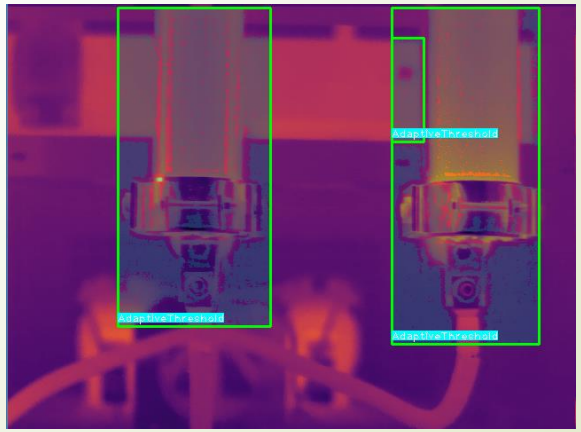
Otsu Algorithm

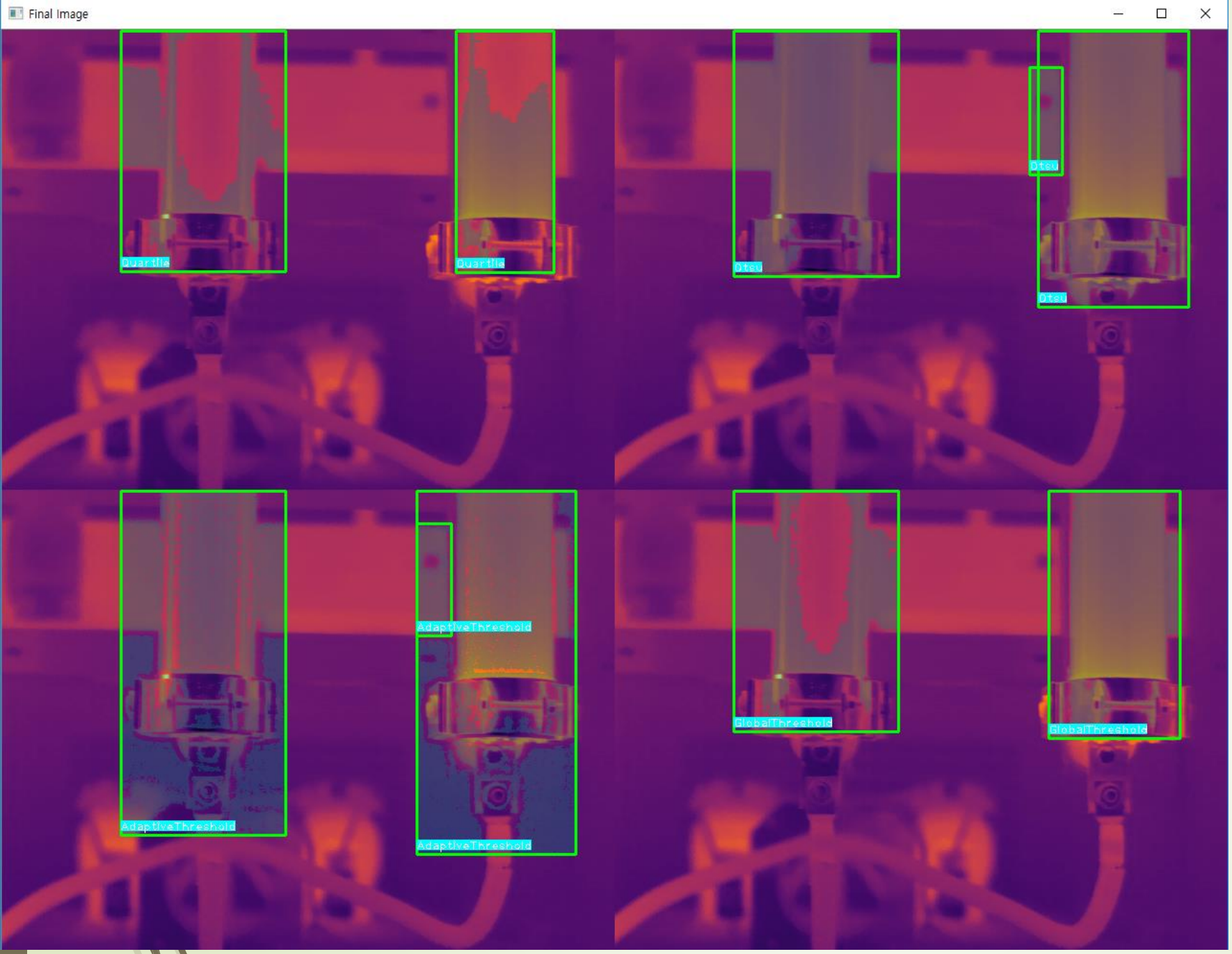


Global Thresholding Algorithm



Adaptive Thresholding Algorithm




Final Image
Quartile
Quartile
Otsu
Otsu
Otsu
AdaptiveThreshold
AdaptiveThreshold
AdaptiveThreshold
GlobalThreshold
GlobalThreshold

Thanks !!